

AB RENEWABLE ENERGY ENGINEERING

Engr. Alexis T. Belonio, MS, ABE, ASEAN Engineer
*Freelance Consultant and Technology Developer; Former
Senior Research Fellow (PhilRice) and
Associate Professor (Central Philippine University)
Member: PSABE, ASABE, ASME*

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SOLAR PHOTOVOLTAIC

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1. Average power density of solar heat striking the earth surface.

- a. 1 kW/m²
- ☒ b. 0.89 kW/m²
- c. 0.58 kW/m²
- d. None of the above

2. Device that generates electricity from the sunlight.

- a. Windpump
- ☒ b. Photovoltaic
- c. Generator
- d. None of the above

3. Component of a solar electric system that converts sunlight to electricity is the ____.

- a. PV panel
- b. solar cell
- c. solar module
- ☒ d. All of the above

4. Solar battery is rated in terms of ____.

- a. dimension of the battery
- b. wattage
- ☒ c. ampere-hour
- d. None of the above

5. Capacity required for a 12volt solar panel system is 73.6 watt-hour. What is the ampere-hour capacity of the battery needed for the system?

- ☒ a. 6.13 ampere-hour
- b. 4.05 ampere-hour
- c. 7.23 ampere-hour
- d. None of the above

6. Factor affecting solar radiation.

- a. Cloudiness
- b. Overcasting
- c. Shade
- ☒ d. All of the above

↑ V = series
↑ X = parallel

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7. Place located near the equator will receive ____ solar radiation than those which are far from it.

- a. less
- b. the same
- ☒ c. more
- d. None of the above

8. If 12volt solar module is required, two 12volt modules must be connected in ____.

- a. series
- ☒ b. parallel
- c. perpendicular
- d. None of the above

9. More electricity can be produced by a solar cell if ____.

- a. the temperature at the collector is higher
- b. there is more cloud in the atmosphere
- ☒ c. there is more light that can be received by the collector
- d. All of the above

10. Suitable heat collector for solar cell.

- a. High-temperature flat-plate collector
- b. Low-temperature parabolic solar collector
- c. Air-inflated type collector
- ☒ d. None of the above

11. Solar modules are usually rated in ____ which indicates the amount of power generated under 1 kW/m² rated condition at 25°C.

- a. watts
- b. volt-ampere
- ☒ c. peak watt
- d. None of the above

12. Approximate size of a solar cell that has the capacity to produce 1 peak watt.

- ☒ a. 10 cm x 10 cm
- b. 5 cm x 10 cm
- c. 10 cm x 20 cm
- d. None of the above

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13. Which of the following does not belong to the PV solar system?

- a. PV Panel or module
- b. Battery System and Charge Controller
- c. Inverter
- d. Load (appliances)
- e. All of the above
- ☒ f. None of the above

14. Device used to convert AC voltage of the solar PV System to DC voltage.

- a. Inverter
- b. Converter
- c. PV panel
- ☒ d. None of the above

DC - AC

15. Collection of solar panels.

- a. Solar modules
- b. Solar cells
- ☒ c. Solar Array
- d. None of the above

16. Which of the following statements is true?

- a. The intensity of solar radiation increases as the distance of the location from the equator increases.
- ☒ b. The intensity of solar radiation decreases as the distance of the location from the equator increases.
- c. The intensity of solar radiation does not change with the distance of the location from the equator.
- d. None of the above

17. When installing a solar panel, the degree of its inclination is usually based on the ____ of the location.

- a. longitude
- ☒ b. latitude
- c. elevation
- d. None of the above

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18. A stand-alone system is ____.

- a. connected to the grid
- ☒ b. not connected to the grid
- c. connected or disconnected to the grid
- d. None of the above

19. Lead-acid battery basically has ____ volts per cell.

- a. 4
- b. 3
- ☒ c. 2
- d. None of the above

20. Solar energy device that generates electricity from the sunlight.

- ☒ a. Photovoltaic cell
- b. Solar collector
- c. Solar generator
- d. None of the above

21. Power output of a solar cell ____ with the increase in temperature .

- a. increases
- ☒ b. decreases
- c. remains the same
- d. None of the above

22. Efficiency of commonly used mono crystalline silicon solar module.

- ☒ a. 14%
- b. 20%
- c. 32%
- d. None of the above

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GOD BLESS !!!

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